Answers on
Booklet - 3

Taken from OCR Physics A past papers

Abbreviations, annotations and conventions used in the Mark Scheme	/	=	alternative and acceptable answers for the same marking point
	;	=	separates marking points
	NOT	=	answers which are not worthy of credit
	( )	=	words which are not essential to gain credit
	____	=	(underlining) key words which **must** be used to gain credit
	ecf	=	error carried forward

1.

(a) weight = 28 x 9.8 / mg — C1
= 270 (N) (274.4) — A1
(using g = 10 then –1)

(b) a completed triangle drawn with correct orientation — B1
at least two labels for triangle with correct directions given — B1

calculation:	scale diagram:	
force P / weight = tan 35	scale given	C1
force P =192 (N)	185 to 200 (N)	A1

(c) tension is greater — B1
(reference to triangle) tension force would be greater (longer) as the holding force P would be larger (longer) for greater angle / larger value needed so vertical component still balances the weight — B1

2.

(a) weight / gravitational force — B1
tension — B1

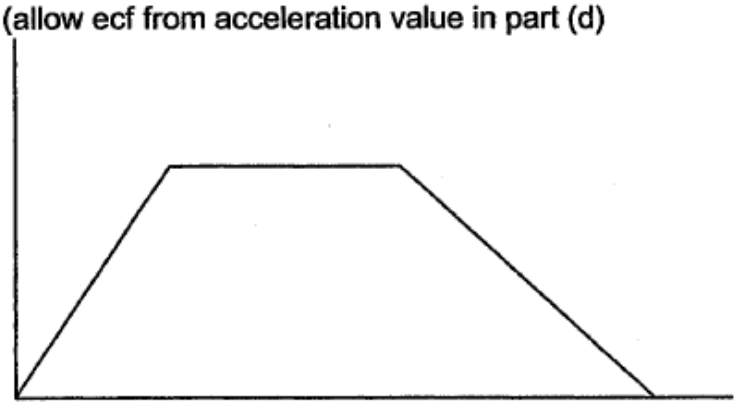
(b) W = (55 + 75) x 9.81 = 1275 (N) (allow 1280) — A1

(c)(i) F = ma — C1
T - 1275 = (130) x 0.55
T = 1346.5 (N) (allow 1350) — A1

(ii) T – W = 0 — C1
T = W = 1275 (N) — A1

(d) T - W = ma
1240 – 1275 = 130 x a — C1
a = (-) 0.27 m s^{-2} — A1

(e) line with constant positive gradient from origin — B1
horizontal line (above zero) — B1
line of constant negative gradient at end taking more time than the first section — B1
(allow ecf from acceleration value in part (d)



3.

the force that causes a mass of one kg to accelerate at 1 m s^{-2}	**B1**
point where (all) the weight / mass of the body	M1
seems appears to act	A1

4.

(a)	density = mass / volume	B1
(b)(i)	density = 45 / 50 x 600 x 600 (x 10^{-9})	C1
	= 2500 and kgm^{-3}	A1
	(or equivalent answer in kg mm^{-3} or kg cm^{-3})	
(ii)	pressure = force / area	C1
	weight = 45 x 9.8	C1
	area = 600 x 50 (x 10^{-6})	C1
	= 14700 / 15000 (Pa)	A1
	(Bold 15000 (Pa) scores ¾ / Bold 14700 (Pa) scores 4/4)	
(iii)	same area on the ground	B1
	same total weight	B1
	same pressure (therefore same damage/or other suitable comment)	B1

5.

(a)(i)	$p = F / A$	M1
	where F is force <u>normal</u> to the cross-sectional area A	A1
(ii)	Pa / Nm^{-2}	B1
(b)(i)	$p = 28 / 8$	C1
	$3.5\ N\ cm^{-2}$ / 35000 Pa	A1
(ii)	$F = p \times 40$	C1
	140 (N)	A1
(iii)	greater force applied to master cylinder / master cylinder smaller (area) / slave cylinder bigger (area) **(any two)**	B2

6.

(a)	**using** $v = u + at$ with $u = 0$ (ie $330 = 0 + 9.8t$) $\checkmark_m$ $t = 330/9.8$ $\checkmark_r$ $= 33.7$ $\checkmark_e$ (s) (**calculator** value shown)	3	3 marking points
(b)(i)	1 distance (travelled) / displacement / height dropped ✓ 2 acceleration ✓	2	
(ii)	($t = 0$ to $t = t_1$) accelerates at decreasing rate ✓ OAW ($t = t_1$ to $t = t_2$) deceleration ✓ ($t = t_2$ to $t = t_3$) constant velocity ✓ ('terminal' velocity insufficient)	3	velocity increases at a decreasing rate velocity decreases constant velocity

7.

(i)	(when a system is in equilibrium) the sum of the anticlockwise moments equals the sum of the clockwise moments (about the same pivot)	B1
(ii)	sum of the forces equals zero / resultant force is zero	B1
	sum of the moments equals zero	B1

8.

(a)	$P = F / A$	C1
	$F = 1.0 \times 10^5 \times 0.25 = 25000$ (N)	A1
(b)	$\rho = m / V$	C1
	$m = 800 \times 0.25 \times 0.75 = 150$	C1
	$W = 1470$ (N)	A1
(c)	26470 (N)	B1

9.

(a)	for a complete set of measurements required, <u>consistent</u> with (b)(ii). ✓✓	2	Deduct 1 mark for any omissions. Mark parts (a) and (b)(i) together
(b)(i)	for explaining each method of taking the specific measurements and stating appropriate measuring instruments ✓✓✓✓	2 2	
(ii)	for logical explanation as to how measurements used to calculate 'a' ✓✓✓	3	F=ma arguments score zero in this section but should be given credit elsewhere within the question.
(c)(i)	Credit two factors (reaction time/ parallax/etc as appropriate to method) which would affect the accuracy of the **method**	2	not friction …… as question refers to this motion
(ii)	relevant suggestion related to (c)(i) ✓ explained ✓	2	

10.

(i)	force drawn vertically upwards at plunger	B1
	force drawn vertically at H	B1
(ii)	20 x 500 / force on Plunger x 120 (one correct moment stated)	B1
	Plunger force x 120 = (20 x 500)	B1
	Plunger force = 83(.3) (N)	A0
:)(i)	pressure = force / area	
	= 83 / 4 x 10^{-3}	C1
	= 20800 (Pa)	A1
(ii)	decrease area of plunger / decrease distance H to plunger / increase F / increase length of arm	B2 MAX 2

11.

(a)	(i)	**Moment is the force x the perpendicular distance from (the line of action of) the force to the pivot/ point (missing perpendicular –1, missing from the force to the pivot / point –1)**	**B2**
	(ii)	**Torque of a couple: one of the forces x perpendicular distance between (the lines of action of) the forces**	**B1**
(b)	(i) 1	**3600 x 1.0 = X x 2.5**	**C2**
		one mark for one correct moment, one mark for the second correct moment and equated to first moment	**A0**
	2	**X = 1440 (N)**	**C1**
		Y = 3600 - 1440 or 3600 x 1.5 = Y x 2.5	**A1**
	(ii)	**= 2160 (N)**	**B1**
		Not a couple as forces are not equal	**B1**
	(iii)	**and not in opposite directions / the forces are in the same direction**	**C1**
		P = F / A	**B1**
		= 1440 / 2.3 x 10^{-2}	**B1**
		= 62609 (6.3 x 10^{4}) unit Pa or N m^{-2}	

12.

(a)(i)	$v^2 = 0 + 2 \times 9.8(1) \times 30$	C1
	$v = 24.(3)$ $(m\ s^{-1})$ (-1 if g = 10 is used ,once only on the paper) (zero scored if s = 36 m is used)	A1
(ii)	$s = ut + ½\ at^2$ or $v = u + at$ or $s = (u+v)t / 2$	
	$30 = 0 + ½ \times 9.8(1) \times t^2$ $t = 24.3 / 9.8$ $t = 2 \times 30/24.3$	C1
	$t = 2.5$ (s)	A1
(b)	In the air: weight / force due to gravity (allow air resistance if included as well)	B1
	(Hence) constant acceleration / acceleration at 9.8 $m\ s^{-2}$ (allow reduced acceleration / terminal velocity if air resistance included)	B1
	In water: weight and (large) fluid resistance / upthrust / buoyancy	B1
	Hence deceleration / slows down	B1

13.

(a)	(i)		$s = ut + ½\ a\ t^2$ $t^2 = (400 \times 2) / 3.9$ $t = 14.3$ (s) allow 14 (s) (2 sf)	C1 A1	Allow any subject Do not allow 14.0
	(ii)		$v^2 = u^2 + 2as$ $v = (2 \times 3.9 \times 400)^{1/2} = 55.9$ $(m\ s^{-1})$ allow 55.8	A1	Allow: $s = (u + v) / t$ or $v = u + at$ but note possible time ecf from part (a)(i)
	(iii)		<s> = 100 / 1.82 = 54.9 $(m\ s^{-1})$	A1	Allow: 55 but not 55.0
(b)			methods of reducing drag/streamlining: eg streamlined helmet, tight fitting clothing to reduce drag methods of increasing acceleration/force: use of poles, 'speed skating' (at start) push off ground	B1	Require what is done and the reason for doing it
			method of reducing friction due to the skis e.g waxing skis / longer/ wider skis	B1	
				B1 MAX 2	

14.

(i)	acceleration = 13 / 20 or gradient attempted	C1
	= 0.65 (m s^{-2}) ± 0.01	A1
(ii)	force = ma / 1200 x 0.65 ecf (b)(i)	C1
	= 780 (N)	A1
(iii)	force = 400 x 0.65 ecf (b)(i)	C1
	= 260 (N)	A1

(i)	(gradient is less hence) acceleration is less / reaches terminal velocity	B1
(ii)	resultant force is less / resistive forces are increasing / driver eases off the accelerator / climbing a hill	B1